

3 (a) (i) Explain what is meant by the *internal energy* of a system.

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..... [1]

(ii) Write the equation representing the first law of thermodynamics, stating the meaning of each of the symbols clearly.

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..... [2]

(iii) Explain why the heat capacity of a fixed mass of gas at constant pressure is greater than its heat capacity at constant volume.

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..... [3]

- (b) A styrofoam cup of negligible heat capacity contains 500 g of water at 90 °C. 100 g of ice at 0 °C is then added to the water.

Data for the specific latent heat of fusion and specific heat capacity of water are shown in Fig. 3.1.

specific latent heat of fusion of water / J kg ⁻¹	3.34×10^5
specific heat capacity of water / J kg ⁻¹ K ⁻¹	4200

Fig. 3.1

Determine the equilibrium temperature of the mixture, assuming no heat loss to the surroundings.

equilibrium temperature = °C [2]

4 Fig. 4.1 shows a block of mass 0.400 kg attached to a light spring with spring constant